

# DRAFT

## CMS Paper

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### Search for long-lived particles decaying to photons in decays of the Higgs boson at $\sqrt{s} = 13$ TeV

The CMS Collaboration

#### Abstract

A search for long-lived particles decaying into photons in rare decays of the Higgs boson is presented. The search uses proton-proton collision data from the CMS detector at the CERN LHC collected between 2016 and 2018, corresponding to an integrated luminosity of  $138 \text{ fb}^{-1}$  at a center-of-mass energy of 13 TeV. A novel technique is introduced that features a kinematic fit to estimate the position of the decay of the hypothetical long-lived particle using the CMS electromagnetic calorimeter, which provides excellent position and energy resolution but lacks longitudinal segmentation. No significant excess above the background prediction is observed. The results are interpreted as upper limits on the branching fraction of the Higgs boson to long-lived particles as a function of the hypothetical lifetime of the long-lived candidate and its branching fraction to photons.

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# 1 Introduction

The discovery of the Higgs boson by the ATLAS and CMS Collaborations in 2012 initiated a precision measurement program aiming to understand the compatibility of the newly discovered particle with the Higgs boson predicted by the standard model (SM). While most measurements performed until today agree with the SM predictions, the constraints of the invisible and beyond the standard model branching fractions are still compatible with the existence of rare Higgs boson decays to new particles, predicted by many theoretical models [1–6].

This letter presents a search for rare Higgs boson decays to long-lived particles (LLPs) that decay to photons. To allow decays of the hypothetical long-lived candidate  $\Phi$  to other particles with significant branching fraction, we require that at least one  $\Phi$  candidate decayed to photons, targeting the  $H \rightarrow \Phi\Phi \rightarrow \gamma\gamma + X$  final state. An efficient online event selection (trigger) is achieved by targeting associated production of Higgs bosons with a W or Z boson decaying to leptons. The characteristic experimental signature of the LLP can be used to reduce backgrounds by reconstructing the decay distance of the LLP candidate from the interaction point. While the CMS crystal electromagnetic calorimeter provides precise measurements of the position and energy of the photon clusters, it does not support longitudinal segmentation and pointing back to the interaction point. A novel technique is introduced where for each mass of the hypothetical long-lived particle  $\Phi$ , a kinematic reconstruction is performed which uses the position and the energy of the photons in the calorimeter. This kinematic reconstruction method is used to estimate the distance of the  $\Phi$  decay from the interaction point with a precision better than 5 cm. The search is sensitive to vertex displacements up to about 90 cm from the beam line. Previous studies on similar final states have been published by the ATLAS Collaboration [7, 8], including a search for Higgs decays to LLPs with each of them decaying to two photons.

## 2 The CMS detector

The CMS apparatus [9, 10] is a multipurpose, nearly hermetic detector, designed to trigger on [11–13] and identify electrons, muons, photons, and hadrons [14–16]. Its central feature is a superconducting solenoid of an internal diameter of 6 m, providing a magnetic field of 3.8 T. Within the solenoid volume are a silicon pixel and strip tracker, a lead tungstate crystal electromagnetic calorimeter (ECAL), and a brass and scintillator hadron calorimeter (HCAL), each composed of a barrel and two endcap sections. Forward calorimeters extend the pseudorapidity coverage provided by the barrel and endcap detectors. Muons are reconstructed using gas-ionization detectors embedded in the steel flux-return yoke outside the solenoid.

In the barrel section of the ECAL, an energy resolution of about 1% is achieved for unconverted or late-converting photons in the tens of GeV energy range. The energy resolution of the remaining barrel photons is about 1.3% up to the pseudorapidity range of  $|\eta| = 1$ , changing to about 2.5% at  $|\eta| = 1.4$ . In the endcaps, the energy resolution is about 2.5% for unconverted or late-converting photons, and between 3 and 4% for the other ones [14].

The electron momentum is estimated by combining the energy measurements in the ECAL with the momentum measurement in the tracker. The momentum resolution for electrons with transverse momentum  $p_T \approx 45$  GeV from  $Z \rightarrow ee$  decays ranges from 1.6 to 5%. It is generally better in the barrel region than in the endcaps, and also depends on the bremsstrahlung energy emitted by the electron as it traverses the material in front of the ECAL [14, 17].

Muons are measured in the pseudorapidity range  $|\eta| < 2.4$ , with detection planes made using

three technologies: drift tubes, cathode strip chambers, and resistive plate chambers. The single muon trigger efficiency exceeds 90% over the full  $\eta$  range, and the efficiency to reconstruct and identify muons is greater than 96%. Matching muons to tracks measured in the silicon tracker results in a relative transverse momentum resolution, for muons with  $p_T$  up to 100 GeV, of 1% in the barrel and 3% in the endcaps. The  $p_T$  resolution in the barrel is better than 7% for muons with  $p_T$  up to 1 TeV [15].

Events of interest are selected using a two-tiered trigger system. The first level (L1T), composed of custom hardware processors, uses information from the calorimeters and muon detectors to select events at a rate of around 100 kHz within a fixed latency of 4  $\mu$ s. The second level, known as the high-level trigger (HLT), consists of a farm of processors running a version of the full event reconstruction software optimized for fast processing, and reduces the event rate to a few kHz before data storage.

A more detailed description of the CMS detector, together with a definition of the coordinate system used and the relevant kinematic variables, can be found in Refs. [9, 10].

### 3 Simulated signal and data samples

This search uses a dataset of proton-proton collisions at the LHC collected from 2016–2018 at  $\sqrt{s} = 13$  TeV, corresponding to an integrated luminosity of 138  $\text{fb}^{-1}$ . The data are selected using single-lepton triggers that require at least one well-reconstructed electron or muon in order to capture events with leptonic decays of the W or Z boson produced in association with the SM Higgs boson.

The primary hard scattering interactions producing the Higgs boson and associated vector boson are created with Monte Carlo (MC) simulation using POWHEG BOX v2 [18–20]. These processes include both quark-induced W, Z, and H production calculated to next-to-leading order (NLO) in quantum chromodynamics (QCD) and electroweak (EW) corrections as well as gluon-induced Z+H production calculated to leading order (LO) in QCD and EW corrections [21]. As the signal is assumed to be produced from decays of SM Higgs bosons and the event selection targets leptonic decays of the W, we expect signal contribution from ttH production modes with semileptonic  $t\bar{t}$  decays, where one top quark decays leptonically and the other decays hadronically. To account for this contribution, signal samples with ttH Higgs boson production with semileptonic  $t\bar{t}$  decays are generated with cross sections calculated at NLO in QCD corrections [22].

The Higgs boson is forced to decay to two ( $\Phi$ ) candidates. Since the event selection efficiency depends on the branching fraction of  $\Phi$  to photons, two signal samples are generated: one with a branching fraction of  $\Phi$  to two photons of 50% (and the remainder to d quarks), and another sample with 100% branching fraction to photons. The event selection efficiency for any hypothetical branching fraction of the  $\Phi$  to photons can then be determined by these two samples, which are generated with a  $\Phi$  mass range of 15 to 55 GeV, and proper lifetimes ( $c\tau$ ) between 0 and 1000 mm. The analysis sensitivity degrades substantially at lower masses and/or longer lifetimes, since the photons tend to merge into a single electromagnetic cluster.

The background is estimated with a data-driven method; however, simulated background samples are still used to validate the methods and tune the analysis selection criteria. The main background contributions originate from Drell-Yan or W boson production, which were simulated to NLO in QCD using the MADGRAPH5\_aMC@NLO generator [23] with 0, 1, and 2 additional jets, and production of top quark pairs that were simulated using MADGRAPH [23, 24].

All simulated samples were produced with parton distribution functions (PDFs) from NNPDF3.1 [25] calculated at next-to-next-to-leading-order (NNLO). The hadronization was generated with PYTHIA 8.240 [26, 27] using the “TuneCP5” tune [28]. Additional proton-proton interactions within the same or nearby bunch crossing (pileup) were superimposed and events were weighted to reproduce the pileup distribution as measured in data. All samples are normalized to the measured integrated luminosity using cross sections and event weights derived using the latest theoretical calculations.

## 4 Object selection and event categorization

Events with leptons are collected by the L1T and HLT single-lepton triggers. To pass the trigger, muon candidates are required to have  $p_T > 24$  GeV in the 2016 and 2018 datasets, and  $p_T > 27$  GeV in the 2017 dataset. Electron candidates are required to have  $p_T > 32$  GeV and pass the “tight” online selection working point. Both flavors of lepton candidates should be isolated in the detector.

In the offline analysis, electrons, muons, and photons are variously identified with a set of “loose” or “tight” criteria for a tradeoff between efficiency and purity.

Loose muon candidates are selected by requiring that they have  $p_T > 5$  GeV,  $|\eta| < 2.4$ , and pass loose identification and isolation requirements. To ensure that the muons originate from the primary vertex, we require that the transverse displacement of the silicon track from the beamspot is  $d_{xy} < 0.2$  cm and the displacement along the  $z$  axis is  $d_z < 0.5$  cm. Tight muon candidates are defined as a subset of loose muons with the additional requirement that they pass “tight” identification and isolation requirements.

Loose electrons are required to have  $p_T > 5$  GeV. Their reconstructed electromagnetic shower should be reconstructed within the barrel or the endcap region of the ECAL by requiring  $|\eta| < 1.44$  or  $1.57 < |\eta| < 2.5$ . The electrons should pass the loose identification and isolation requirements. Their transverse and longitudinal distance from the primary vertex should be smaller than 0.2 cm and they should pass a set of selection requirements that veto electrons originating from photon conversions in the tracker material. Tight electron candidates are also selected as a subset of loose electrons, by requiring that they pass tight identification and isolation requirements.

Photon candidates are required to have  $p_T > 25$  GeV and their reconstructed electromagnetic shower should also satisfy  $|\eta| < 1.44$  or  $1.57 < |\eta| < 2.5$ . To reject electrons misidentified as photons, we apply a veto on the presence of pixel track seeds in the vicinity of the photon. The photons are required to pass just loose isolation criteria to be qualified as loose photons and both loose isolation and identification requirements to be qualified as tight photons. For large photon displacements, the tight photon identification efficiency is mildly affected by the change of the shower shape when photons enter the ECAL crystals (that are projected to the detector origin) at an angle. The tight identification efficiency for signal was estimated from simulation and is about 90% for displacements below 40 cm, and drops to about 75% for displacements of 90 cm from the beam line. Since reconstructing both photons near the calorimeter is unlikely due to merging, the efficiency loss from the cluster shape has a negligible impact on the overall sensitivity.

The events are initially selected by requiring a reconstructed  $W$  or a  $Z$  boson.  $Z$  bosons are identified by requiring the presence of a pair of same-flavor and opposite-sign tight leptons with an invariant mass of  $70 < m_{\ell\ell} < 110$  GeV.  $W$  bosons are identified by selecting events

with only one tight lepton. The muon candidates used to select W and Z events should have  $p_T > 25$  GeV in 2016 and 2018, and  $p_T > 28$  GeV in 2017, while the respective electron candidates should have  $p_T > 35$  GeV. The lepton candidates with the largest  $p_T$  must be matched to the respective triggered candidates.

We require that photons do not spatially overlap with any loose leptons by requiring that  $[\Delta\eta(\ell, \gamma)/0.4]^2 + [\Delta\phi(\ell, \gamma)/0.5]^2 > 1$ , which takes into account the asymmetric cluster shape in the ECAL originating from the large magnetic field of the solenoid. Finally we require that the events have at least two photons that pass the criteria described above.

The selected events are initially separated into three categories based on the year of data taking (2016, 2017, 2018). For each year, we split the events based on the reconstruction of a W or Z boson candidate. Finally, for each vector boson, the events are separated further in muon and electron categories based on the lepton flavor, resulting in twelve final event categories.

## 5 Reconstruction of the $\Phi$ decay vertex

For each category, the invariant mass of the two selected photons ( $m_{\gamma\gamma}$ ), under the assumption that they originate from the primary vertex, can be used to discriminate between the signal and background. Figure 1 shows the  $m_{\gamma\gamma}$  distribution for signal events under different hypotheses of the  $\Phi$  lifetime for  $m_\Phi = 30$  GeV. We observe that the reconstructed diphoton mass resolution depends strongly on the position of the vertex that is used to define the coordinates of the photon four-vectors, resulting in a broadening and shifting of the distribution.

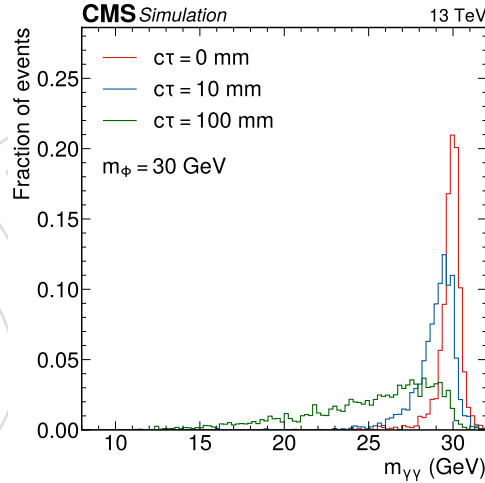


Figure 1: Reconstructed diphoton invariant mass for signal with  $m_\Phi = 30$  GeV for different values of the  $\Phi$  lifetime.

Since we focus on long-lived particles with large lifetimes, we introduce a kinematic reconstruction of the  $\Phi$  decay displacement that is performed on the final set of selected events for each  $m_\Phi$  hypothesis. The mass of the  $\Phi$  decaying to two photons can be estimated by:

$$m_\Phi^2 = 2E_1E_2(1 - \cos\theta), \quad (1)$$

where  $m_\Phi$  is the mass of the  $\Phi$ ,  $E_1$  and  $E_2$  are the energies of the two photons, and  $\theta$  is the angle between the two photons at the point of the  $\Phi$  decay. Under the assumption that the  $\Phi$  is produced at the primary vertex (PV), the PV, secondary  $\Phi$  decay vertex, and ECAL clusters are coplanar.



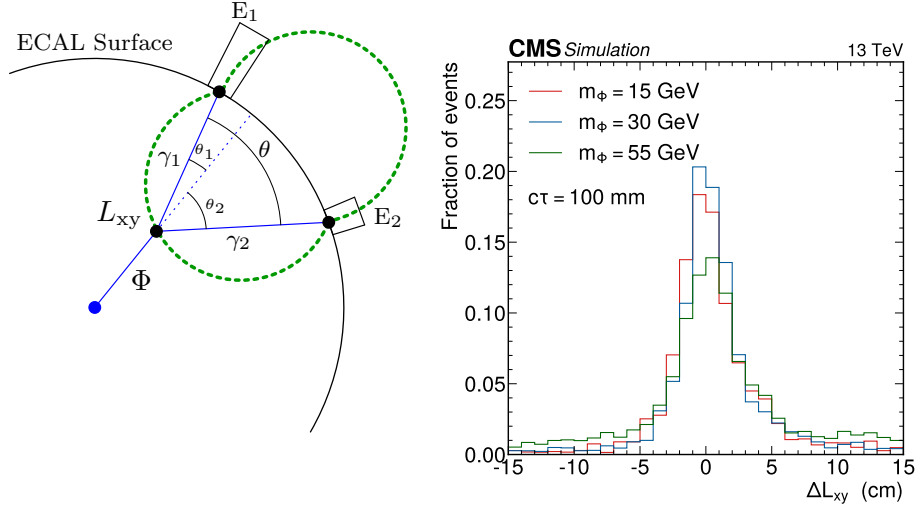


Figure 2: Left: Geometric configuration used to reconstruct the displaced vertex of the diphoton decay. Solid blue lines represent particle trajectories, while the dashed green lines represent the locus of possible decay vertices. Right: Resolution of the reconstructed transverse displacement  $L_{xy}$  in signal events for different values of the  $\Phi$  candidate mass.

The left side of Figure 2 shows the geometric configuration in the plane of the decay. The assumed value of  $m_\Phi$  and the measured energy of the photons define the angle between the two photons ( $\theta$ ) using Eq. (1). The  $\Phi$  decay vertex and the measured position of the two ECAL clusters in the face of the calorimeter are three non-collinear points which can belong to two circles with inscribed angle  $\theta$ . If  $\theta < \pi/2$ , only the major arcs of the two circles satisfy Eq. (1). Alternatively, if  $\theta > \pi/2$ , only the minor arcs represent valid solutions. In either case, the arc from one of the two circles will extend outside the calorimeter, representing a non-physical solution that can be discarded. The remaining arc represents the locus of points containing possible decay vertices of the  $\Phi$ . The angle between the two photons can be divided into the deflection of each photon ( $\theta_1$  and  $\theta_2$  respectively). We exploit the conservation of momentum that requires  $E_1 \sin \theta_1 - E_2 \sin \theta_2 = 0$  to numerically calculate the vertex by sampling points along the physically allowed arc, and choosing the coordinates that best satisfy the conservation of momentum. The final step is to project the solution back to the lab frame, and reconstruct the transverse distance  $L_{xy}$  between the estimated diphoton vertex and the center of the detector. The method is validated using simulated signal samples. When comparing the reconstructed vertex position with the vertex position in simulation, we estimate a resolution better than 5 cm for the  $L_{xy}$  displacement across the whole  $m_\Phi$  and  $L_{xy}$  analysis range as it can be seen on the right side of Figure 2. Using this method, we reinterpret the data events for each different  $m_\Phi$  hypothesis, enabling a signal extraction using the  $L_{xy}$  distribution.

The behavior of background events through this kinematic reconstruction method can be summarized in two distinct cases: If the diphoton invariant mass under the assumption that the decay happened at the PV is larger than the  $m_\Phi$  hypothesis, the vertex is reconstructed behind the interaction point, resulting in negative values of the measured  $L_{xy}$ . These events provide an adjoint region used to validate the employed background estimation methods. On the other hand, if  $m_{\gamma\gamma}$  is smaller than the  $m_\Phi$  hypothesis, the angle  $\theta$  increases promoting the diphoton system to larger displacement. This type of background enters the signal region and needs to be reduced with topological requirements.

## 6 Topological selection criteria

One major contribution of background events in the analysis arises from final-state radiation of photons from nearby leptons. Even if the geometric overlap cleaning removes a large fraction of this background, investigation of simulated events showed that there are rare cases in muon events where a muon will radiate more than 50% of its energy, resulting in a soft muon near the photon. To reduce this source of final-state radiation, we define a new set of muons that are required to pass the soft identification requirements and have  $p_T > 3$  GeV and  $|\eta| < 2.4$ . Then we require that all photons considered in the final selection also satisfy  $\Delta R(\mu, \gamma) = \sqrt{\Delta\eta^2 + \Delta\phi^2} > 0.8$ .

Another background contribution arises from electrons misidentified as photons in Z events. In this case, one electron from the Z boson was misidentified as a photon and entered the event selection as a W candidate event. To reduce this background, we form the invariant mass of the electron and the photon candidates, requiring that both photons satisfy  $|m_{\gamma e} - m_Z| > 10$  GeV. Since sometimes the electron is reconstructed as two photons, we also require that  $|m_{\gamma\gamma e} - m_Z| > 15$  GeV.

Next, we address the background that arises from events with  $m_{\gamma\gamma} < m_\Phi$  that are “promoted” by the kinematic fit to larger displacements, as discussed in Section 5. These events are dominated by soft photons, therefore the kinematic fit attempts to equalize  $m_{\gamma\gamma}$  with  $m_\Phi$ , resulting in a vertex with large displacement. Since these events have photons that are less energetic than the ones in signal events, we reduce this background by requiring that the leading photon has  $p_T > 35$  GeV while the subleading photon has  $p_T > 25$  GeV.

Finally, the magnitude of the vectorial sum of the transverse momentum of the photon pair should be  $p_T(\gamma_1, \gamma_2) > 20$  GeV, since it is expected to be boosted by the Higgs mass and the recoil of the vector boson. Moreover, we require that  $m_{\gamma\gamma} > 4$  GeV, rejecting a large fraction of background events with nearby electromagnetic fluctuations inside jets. In the rare case where more than two photons satisfy the selection requirements, we retain the pair that has a kinematically allowed solution for the vertex reconstruction. In case of ambiguity, we select the pair that satisfies stricter identification requirements, preferring tight photons over loose photons. Finally, in case of all photons passing tight identification, the pair with the largest invariant mass as calculated from the primary vertex is chosen.

The signal region is defined as a region that has a vector boson and two tight photon candidates in the final state. A background extrapolation region, defined by events with one tight and one loose photon, is used to estimate the background in the signal region, as discussed in the following Section. To account for the resolution of the vertex reconstruction, we consider only events with  $L_{xy} > -10$  cm in the signal region.

## 7 Background estimation

As discussed in section 6, the main sources of background arise from photons radiated in the initial and final state, and non-prompt photons from jets with a large electromagnetic energy fraction. Since the number of simulated events is not sufficient to reliably estimate these contributions, a data-driven method is employed, that predicts the number of background events in each  $L_{xy}$  bin from events with loosely selected photons. The method utilizes a background extrapolation factor  $f_\gamma$  that is defined as the ratio between the number of photons that passed tight photon selection, divided by the number of photons that passed loose selection, as defined in Section 4. The extrapolation factor is measured as a function of the photon  $p_T$  and  $\eta$ .



To measure  $f_\gamma$ , we use a disjoint category of events that contain a W boson decaying to leptons, and *only one* loose photon. The requirement of the leptonic W boson decay is introduced to provide a similar environment to the signal region, where there is significant contribution of initial-state and final-state radiation. The extrapolation factor  $f_\gamma$  is measured separately for electron and muon events, and ranges from 2% to 16% for events with muons and between 2.5% and 20% for events with electrons.

To estimate the number of background events in each  $L_{xy}$  bin, we select events with the leading photon passing tight selection criteria and subleading photon passing loose selection criteria, and apply a weight on each event equal to the value of  $f_\gamma$  as a function of the subleading photon  $p_T$  and  $\eta$ . In the categories with Z bosons, we use the measurement of  $f_\gamma$  in the corresponding W categories, since the number of selected Z events was not large enough to enable a measurement of the extrapolation factor with reasonable uncertainty. Since the extrapolation factor is applied to events that belong to a superset of the signal region, we estimated the expected signal contamination with signal yield close to the expected 95% CL upper limit. The signal contamination is well below 10% for the W categories and up to 25% in some bins in the Z categories. In all  $L_{xy}$  bins in all categories, the statistical uncertainty of the estimated background is significantly larger than the expected signal yield.

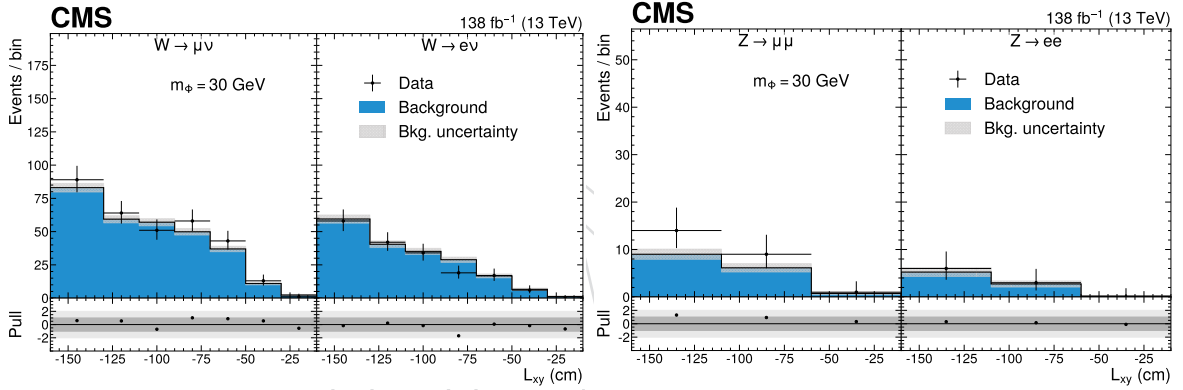


Figure 3: Comparison between data and predicted background events for events with  $L_{xy} < -10$  cm and  $m_{\gamma\gamma} > 62.5$  GeV for W categories (left) and Z categories (right), demonstrating good agreement between the background prediction and data. The impact parameter was calculated with an assumption of  $m_\Phi = 30$  GeV.

The background estimation method is validated in simulation by consistently measuring and applying  $f_\gamma$  in a set of simulated samples from W and Z associated with photons and jets, and events with top quarks. The method is also tested with data, by selecting events passing the final selection that have a negative measured value of  $L_{xy}$ . To ensure that signal events from a different  $m_\Phi$  hypotheses do not enter this validation region, an additional requirement is made, where the raw invariant mass is selected to be above half of the Higgs boson mass. Figure 3 shows the results of this test, using a mass hypothesis of  $m_\Phi = 30$  GeV, and demonstrating good performance of the background estimation method.

## 8 Signal extraction and systematic uncertainties

The branching fraction of the Higgs boson to a pair of  $\Phi$  candidates,  $\mathcal{B}(H \rightarrow \Phi\Phi)$ , is extracted by maximizing a likelihood using  $L_{xy}$  as the statistical estimator. Systematic uncertainties contribute to both signal and background processes, and are introduced in the likelihood as nuisance parameters. Each  $L_{xy}$  bin in each category is considered as a separate Poisson process

with a single contribution for the background events predicted with the method described in Section 7, and separate contributions for each signal process. In the W categories, we consider signal contributions from associated production of the Higgs boson either with a W boson or with a top quark pair. In the Z categories, we consider associated production with a Z boson through both the quark-initiated and the gluon-initiated processes. In both cases, we use the Higgs production cross sections as described in Section 3. The acceptance and the reconstruction efficiency of the final state selection of a vector boson and two photons depends strongly on the branching fraction of the  $\Phi$  candidate to photons,  $\mathcal{B}(\Phi \rightarrow \gamma\gamma)$ , since the probability to reconstruct at least two photons is larger in events where both  $\Phi$  candidates decayed to photons. Therefore, we repeat the signal extraction for each  $\mathcal{B}(\Phi \rightarrow \gamma\gamma)$  hypothesis. The number of expected signal events for each signal process can be interpolated using samples where only one  $\Phi$  candidate decays to photons or both  $\Phi$  candidates decay to photons, respectively, as following:

$$N = \mathcal{L}\sigma\mathcal{B}(\text{H} \rightarrow \Phi\Phi) \left\{ \epsilon_{2\gamma} 2\mathcal{B}(\Phi \rightarrow \gamma\gamma) [1 - \mathcal{B}(\Phi \rightarrow \gamma\gamma)] + \epsilon_{4\gamma} \mathcal{B}(\Phi \rightarrow \gamma\gamma)^2 \right\}, \quad (2)$$

where  $\sigma$  is the Higgs boson production cross section, and  $\epsilon_{2\gamma}$  and  $\epsilon_{4\gamma}$  are the products of the acceptance and efficiency of the complete event selection in events where one  $\Phi$  candidate, or both  $\Phi$  candidates decay to photons respectively. For each hypothesis of  $\mathcal{B}(\Phi \rightarrow \gamma\gamma)$ , the event yield is derived by weighing the two contribution in Eq. 2 by the respective factors that depend on the branching fraction of the  $\Phi$  to photons.

The simulated signal samples are sensitive to the theoretical uncertainties related to the production cross section of the Higgs boson. Scale effects of the QCD,  $\alpha_s$ , and PDF uncertainties are implemented for each production model as calculated in Ref. [29]. The expected signal yield is directly affected by the uncertainty in the total collected luminosity and ranges from 0.73–0.84% across all three data-taking eras [30]. The expected signal yield is also directly influenced by the scale factors of the efficiencies, which are calculated using tag-and-probe analyses to improve the agreement of the reconstruction efficiencies between data and simulation. The uncertainties associated with these scale factors are propagated to calculate their effect on the expected signal yield and are generally on the order of 1% or smaller. The number of signal events in the simulation is small in some  $L_{xy}$  bins and contributes a systematic uncertainty that ranges from 1% to 15%.

The data-driven prediction of the background is subject to uncertainties of the amount of data in the extrapolation region consisting of one tight and one loose photon, and to the statistical uncertainty of the derivation of the extrapolation factor in events with a W boson candidate and a single photon. The uncertainty of the Poisson-weighted events in the extrapolation region is obtained using a Poisson bootstrap process [31]. For each event weight we sample a Poisson distribution with parameter  $\lambda = 1$  a thousand times and we multiply the number of events in each sample with the event weight. Consequently, we numerically calculate the 68% CL from this distribution. This uncertainty ranges from 25% to 75% and is modeled in the signal extraction likelihood with a lognormal distribution. In the case of a single event or no events in the extrapolation region, we model this uncertainty with a gamma function where the extrapolation factor is the single event weight or the average extrapolation factor, respectively. Finally, the uncertainty of the extrapolation factor, derived from the 68% CL of the binomial distribution, and modeled as a lognormal distribution in the likelihood, is smaller than 5%.

## 9 Results

The final distributions of events for each category in the signal region are shown in Figs. 4 and 5 before the likelihood maximization is performed.

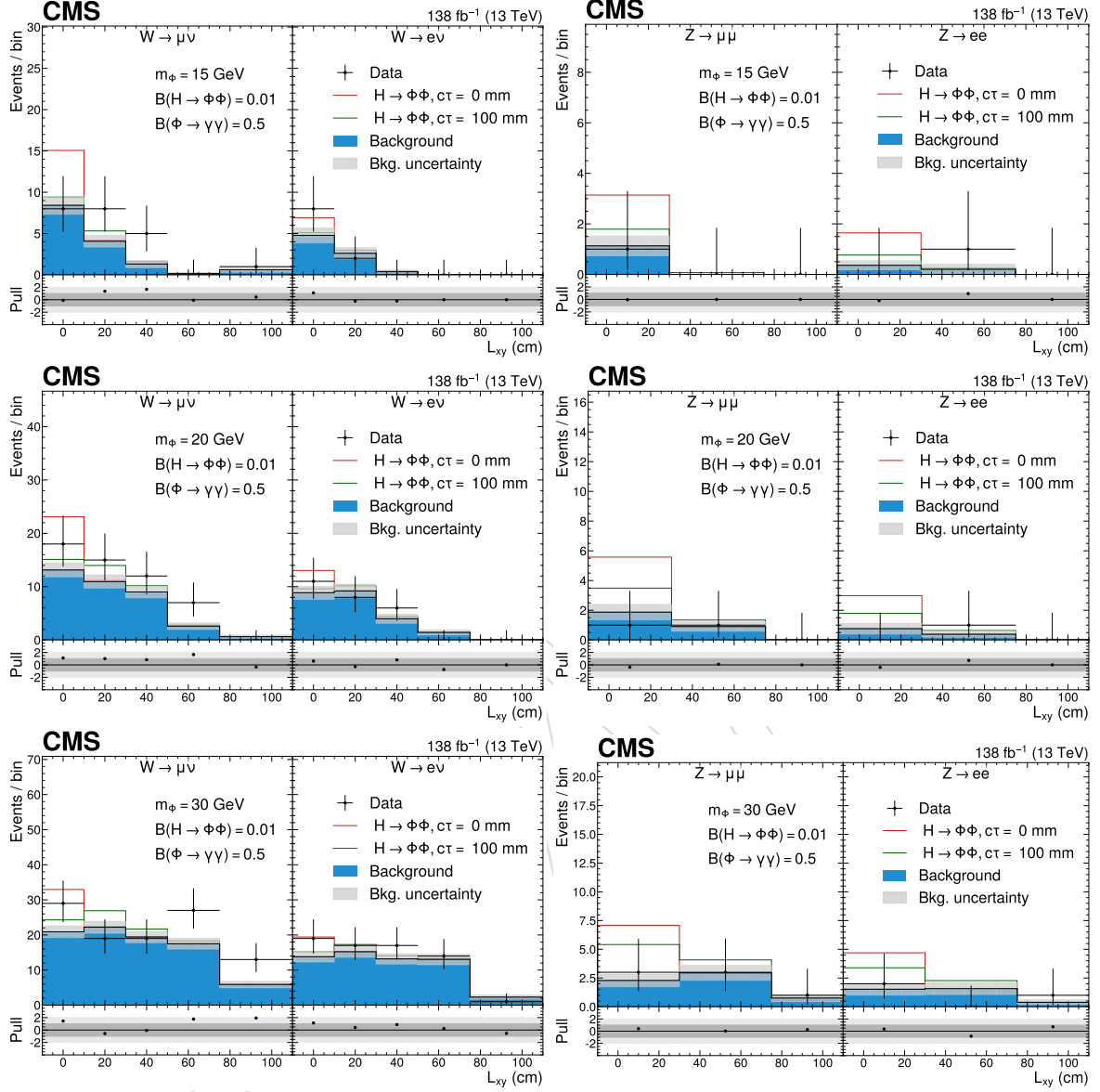


Figure 4: Comparison of data yields with the background prediction for  $m_\Phi$  hypotheses of 15, 20, and 30 GeV for the W categories (left) and the Z categories (right). The distributions from all periods of data taking have been summed together. Expected signal contributions are overlaid for  $c\tau = 0$  mm and  $c\tau = 100$  mm under the assumption of  $B(H \rightarrow \Phi\Phi) = 1\%$  and  $B(\Phi \rightarrow \gamma\gamma) = 50\%$ .

In these distributions, all three years of data-taking are summed together to a single contribution. Since each plot corresponds to a given hypothesis of  $m_\Phi$ , the same events are interpreted many times, giving a different value of  $L_{xy}$  for each  $m_\Phi$  hypothesis. Signal contributions are overlaid for values of the lifetime of  $c\tau = 0$  mm and  $c\tau = 100$  mm. The signal normalization corresponds to a branching fraction of the Higgs boson to  $\Phi$  candidates of 1% and a branching fraction of  $\Phi$  to photons of 50%. The observed number of events and predicted background event yields are in agreement over the whole phase space of the analysis.

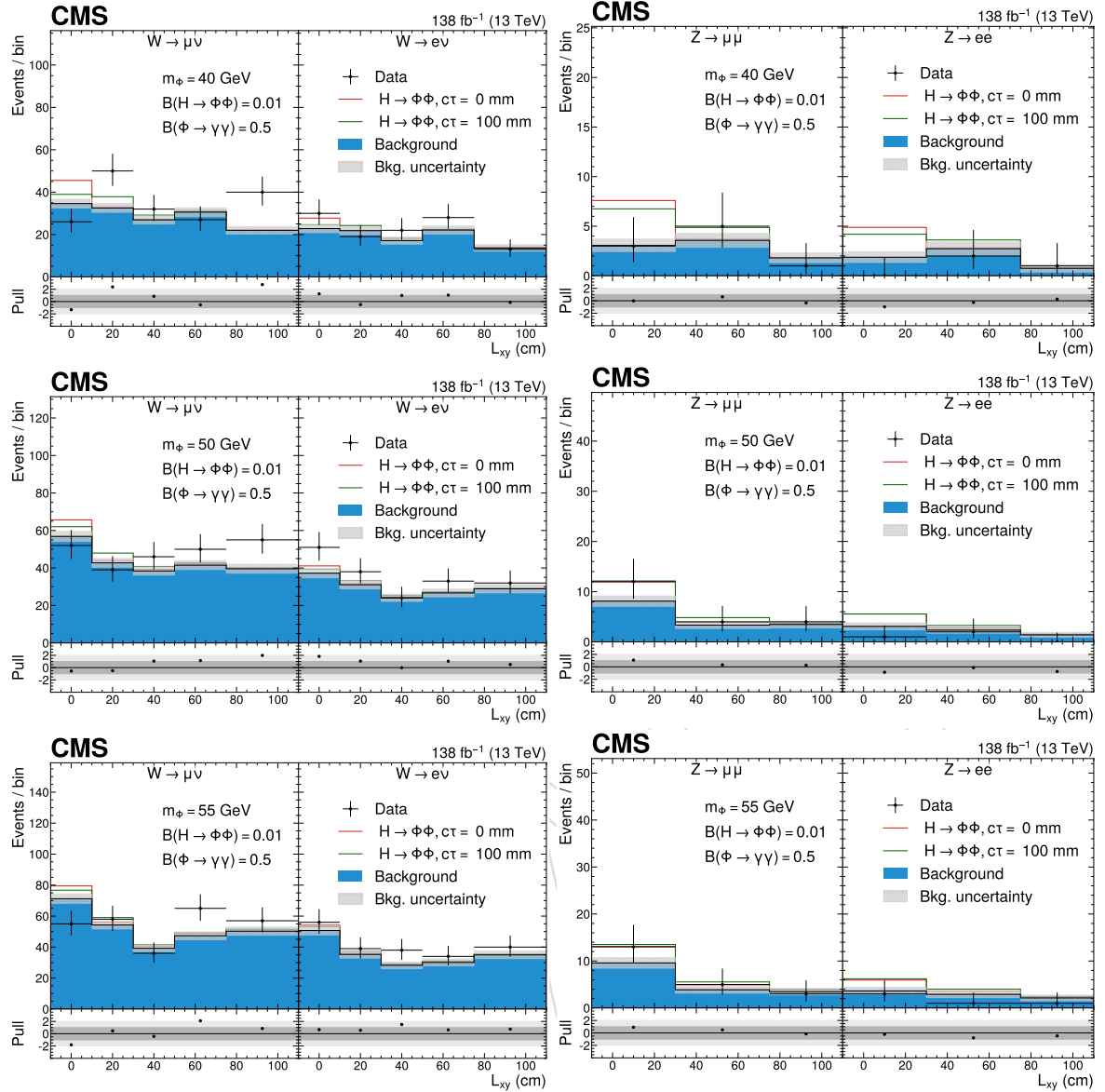


Figure 5: Comparison of data yields with the background prediction for  $m_\Phi$  hypotheses of 40, 50, and 55 GeV for the W categories (left) and the Z categories (right). The distributions from all periods of data taking have been summed together. Expected signal contributions are overlaid for  $c\tau = 0$  mm and  $c\tau = 100$  mm under the assumption of  $\mathcal{B}(H \rightarrow \Phi\Phi) = 1\%$  and  $\mathcal{B}(\Phi \rightarrow \gamma\gamma) = 50\%$ .

The statistical analysis was performed using the CMS statistical analysis tool COMBINE [32]. Since no significant deviation from the background-only hypothesis is observed for any value of  $m_\Phi$  and  $c\tau$ , the results are presented as 95% CL upper limits of the branching fraction of the Higgs boson to  $\Phi$  candidates as a function of  $m_\Phi$  and  $c\tau$ , for different values on the branching fraction of  $\Phi$  to photons. The results are shown in Figure 6 both as a function of lifetime for a fixed  $\mathcal{B}(\Phi \rightarrow \gamma\gamma)$  of 0.05 and as a function of  $\mathcal{B}(\Phi \rightarrow \gamma\gamma)$  for a fixed lifetime of 100 mm. For values of the  $\Phi$  lifetime up to 100 mm, we exclude a branching fraction of the Higgs boson to LLPs of 10% or lower. Based on the expected limits for  $c\tau = 1000$  mm, the analysis is not sensitive for very large lifetimes, since the decay often happens outside or very near the ECAL, resulting in low event selection efficiency. We observe that the analysis is most sensitive for

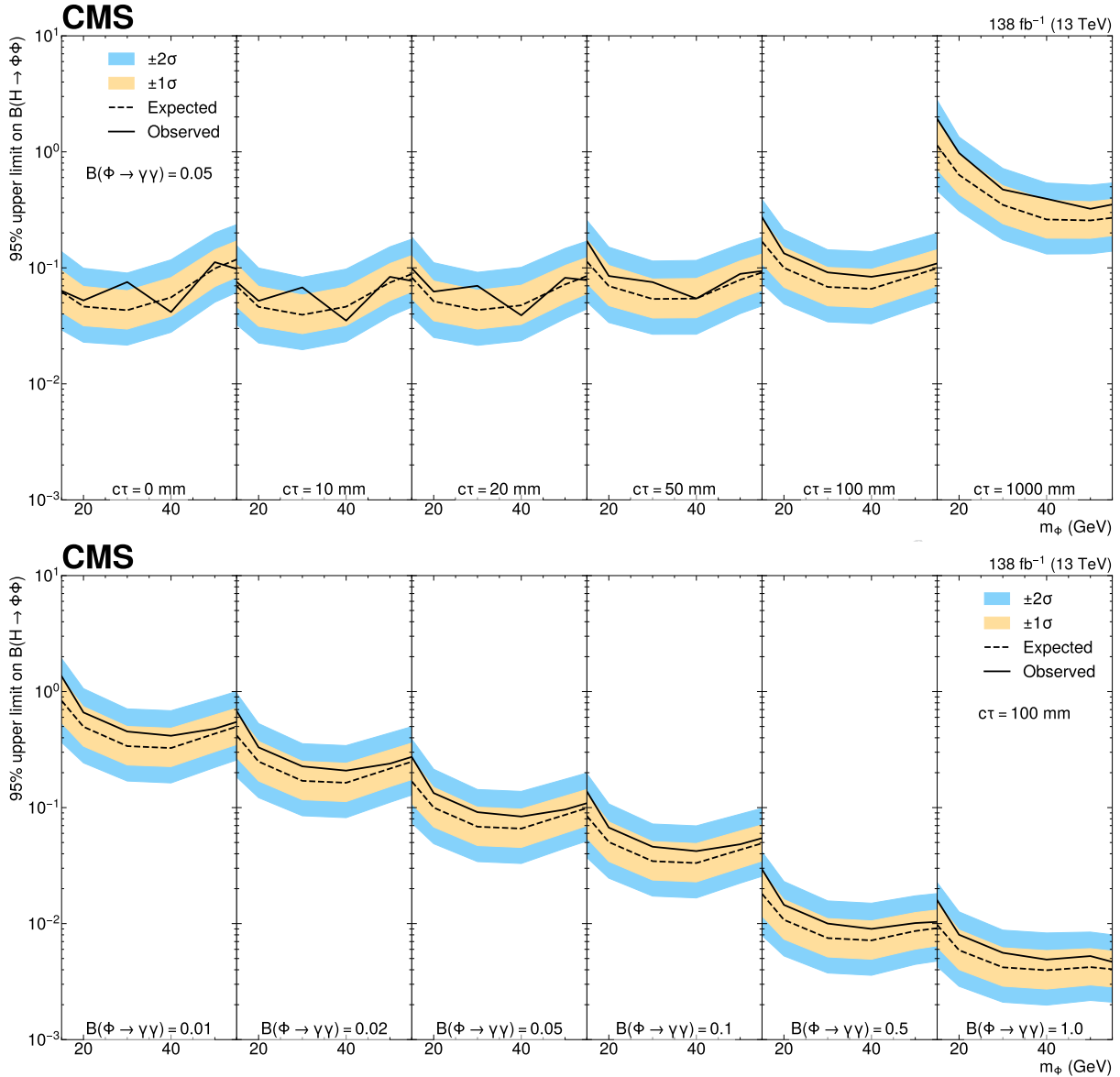


Figure 6: Summarized 95% CL upper limits on the branching fraction of the Higgs boson to the new scalar particle  $\Phi$ . Top: Upper limits presented as a function of  $m_\Phi$  and lifetime for a value of the branching fraction of  $\Phi$  to photons of 0.05). Bottom: Upper limits presented as a function of  $m_\Phi$  and  $B(\Phi \rightarrow \gamma\gamma)$  for a lifetime of 100 mm.

intermediate masses around 30 GeV, since for lower masses the identification and isolation efficiency is lower, while at higher masses the background is larger as it can also be seen in Figures 4 and 5. The complete set of upper limits for all  $B(\Phi \rightarrow \gamma\gamma)$  and  $c\tau$  hypotheses can be found in [33].

## 10 Summary

A search for rare decays of the SM Higgs boson to LLPs decaying to a pair of photons has been presented, using pp collisions at 13 TeV collected with the CMS detector from 2016–2018, corresponding to an integrated luminosity of 138 fb<sup>-1</sup>. A novel technique using kinematic constraints to geometrically reconstruct the decay vertex of the two photons has been employed

to set upper limits on the branching fraction of the Higgs boson. Expected 95% CL upper limits probe branching fractions of the Higgs boson to  $\Phi$  candidates below 10% for modest branching fraction of the  $\Phi$  candidate to photons. This analysis represents the first search by the CMS experiment for long-lived particle decays to photons in a mass range of 15–55 GeV and proper lifetimes up to 1000 mm.

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